Capstone Project Preliminary Analysis

Name

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Project Title

Portfolio Health and Profitability

Project Objective

The objective of this assessment is to develop an executive level overview of the commercial lending portfolio and evaluate its overall health and profitability. The analysis will examine performance across officers, branches, and the consolidated portfolio to identify strengths, risks, and opportunities for improvement.

Data Collection Source

This dataset consists of live, production level information obtained from a community commercial lending bank. All data was extracted from internal reporting systems prior to anonymization.

Methodology:

To ensure comprehensive coverage of the bank's commercial lending and deposit activity, four internal reports were generated:

- Customer information
- Loan information
- Account information
- Collateral information

Each report was exported into its own Excel file. Before any data could be removed from the Bank's secure environment, all personally identifiable information (PII) was stripped and validated by upper management.

In Excel, unique surrogate identifiers were created for customer names, loan account numbers, deposit account numbers, branch names, and officer names. XLOOKUP functions were used to map original PII to the newly assigned identifiers. While most customer records matched automatically, a subset required manual verification to ensure accuracy.

All columns were standardized to lowercase and renamed for clarity and consistency.

Preprocessing Techniques

The following steps were used to organize, clean, and prepare the dataset for analysis:

Data Organization

Each report was structured into a tabular format with consistent column naming conventions. Related fields were grouped logically to support downstream SQL ingestion.

Workflow

- Export internal reports
- Remove and validate PII
- Assign surrogate identifiers
- Standardize name columns and formats
- Perform data quality checks (missing values, duplicates, data types)
- Prepare files for SQL import

Tools Used

- Microsoft Excel (cleaning, validation, identifier assignment)
- SQL (planned for transformation, modeling, and basic analysis)

Data Cleaning

- Corrected typos and inconsistent spellings during manual verification
- Converted data types (e.g., string > dates, strings > numeric)
- Trimmed and normalized text fields
- Removed all PII columns entirely

Missing Values

Missing values were reviewed to determine whether they represented true nulls, data entry gaps, or system limitations. Decisions regarding replacement or exclusion will be based on analytical impact (e.g., profitability calculations, risk metrics)

Duplicates

Duplicate detection was performed using Excel conditional formatting

- No duplicate line items were found in the Loan or Account files after validating unique account numbers.
- The Customer file contained several duplicates due to multiple relationships or legacy system entries.

Exclusions

Any columns containing sensitive PII were removed entirely. Additional exclusions may be applied once SQL modeling begins (e.g., unused system fields, redundant identifiers).

Findings:

Data Size

TABLE NAME	ROWS	COLUMNS
COLLATERAL	1,408	12
CUSTOMERS	20,246	9
LOANS	2,380	140
ACCOUNTS	11,762	63
TOTALS	35,796	224

Missing Values

Missing values are not present among any dataset. All values that are shown as zero or blank should be shown as such. There is no need to further analyze how to fill data gaps or system limitations.

Duplicates

- No duplicate line items were found in the Loan or Account files after validating unique account numbers.
- The Customer file contained several duplicates due to multiple relationships or legacy system entries. These were resolved to the best of the ability during data cleaning within Excel.

Dates

The dataset includes a wide range of dates covering account opening, maturity, rate review, and payment dates. These dates range from 1998 through 2026.

Additional Findings

Preliminary review of the datasets revealed:

- Inconsistent name conventions for customer information (e.g., last name changed to married name in one aspect of system and not another, business names with inconsistent spacing and punctuation within name)

Data Sets

Only fully cleaned datasets have been provided, as all PII was scrubbed prior to removal from the secured system.

Attach any SQL or Python code in a separate file